

Generation Interconnection Request System Impact Study Report

For Queue Project Q15-0467

Somerville 345 kV
March 22, 2017

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0467. The Interconnection Customer proposes a 354.8 MW wind facility interconnecting at the Somerville 345 kV substation in the Pennsylvania area. The request was received on April 27, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 200 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 8 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$16,476,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: CHASE HILL-RECLAIM 138.0 kV
Contingency: P.T.S-BURWELL 345.0 kV L
Loading: 136.8% (Rating: 210 MVA)
DFAX: 11.3%
Pre-project loading: 117.6%
2. Facility: RECLAIM-CENTERVILLE RD 138.0 kV
Contingency: BURWELL-P.T.S 345.0 kV
Loading: 132.8% (Rating: 233 MVA)
DFAX: 11.3%
Pre-project loading: 115.5%
3. Facility: SAN CARLOS-CENTERVILLE RD 138.0 kV
Contingency: P.T.S-BURWELL 345.0 kV
Loading: 131.8% (Rating: 244 MVA)
DFAX: 11.3%
Pre-project loading: 115.4%
4. Facility: SAN CARLOS-SOMERVILLE 138.0 kV line
Contingency: BURWELL-P.T.S 345.0 kV
Loading: 131.4% (Rating: 252 MVA)
DFAX: 11.3%
Pre-project loading: 115.5%
5. Facility: SOMERVILLE 138/1kV Transformer
Contingency: BURWELL-P.T.S 345.0 kV L
Loading: 124.3% (Rating: 424 MVA)
DFAX: 24.3%
Pre-project loading: 104.0%

6. Facility: WILSON 1&2 345 kV/138 kV Transformer
 Contingency: SOUTHEAST VICKSBURG-ALLEGHENY ENERGY UNITS 3 4 & 5 500.0 kV L
 Loading: 110.8% (Rating: 374 MVA)
 DFAX: 7.8%
 Pre-project loading: 103.4%
7. Facility: WAIAWA-N ALLEGHENY WINDPOWER PROJECT 345.0 kV
 Contingency: MERRET-SUNRISE CENTER 500.0 kV
 Loading: 109.5% (Rating: 1,447 MVA)
 DFAX: 5.7%
 Pre-project loading: 108.2%
8. Facility: THERMAS-KILARC 138.0 kV
 Contingency: CONEMAUGH HYDRO PLANT-THERMAS 138.0 kV
 Loading: 105.0% (Rating: 177 MVA)
 DFAX: 5.1%
 Pre-project loading: 94.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
CHASE HILL-RECLAIM 138.0 kV	Rebuild line with higher-capacity conductor	\$1,084,000	46% (\$494,000)
RECLAIM-CENTERVILLE RD 138.0 kV	Rebuild line with higher-capacity conductor	\$2,020,000	46% (\$921,000)
SAN CARLOS-CENTERVILLE RD 138.0 kV	Rebuild line with higher-capacity conductor	\$722,000	46% (\$329,000)
SAN CARLOS-SOMERVILLE 138.0 kV	Rebuild line with higher-capacity conductor	\$1,520,000	46% (\$693,000)
SOMERVILLE 138/1kV Transformer	Replace transformer with higher-rated unit	\$5,547,000	65% (\$3,620,000)
WILSON 1&2 SUBSTATION 345 kV/138 kV Auto	Replace transformer with higher-rated unit	\$14,096,000	60% (\$8,458,000)
N ALLEGHENY WINDPOWER PROJECT-WAHIWA 345.0 kV	Reconductor line	\$11,091,000	8% (\$884,000)
KILARC-THERMAS 138.0 kV	Reconductor line	\$1,550,000	69% (\$1,077,000)
	Total allocated to project		\$16,476,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.